2.4.1.1 Fundamental principles of operation

- *Remote control principle*: The first-generation drones operated through basic radio frequency (RF) control, where an operator transmitted commands using a ground-based controller. The aircraft's onboard receiver interpreted these signals and activated servomechanisms to adjust control surfaces such as the rudder, elevator, and ailerons. This established the foundation for modern UAV communication systems. •• ••
- *Aerodynamic lift and stability*: Like conventional aircraft, these drones relied on fixed-wing aerodynamics. Lift was generated as air moved across the wings, while the tail and control surfaces provided balance and directional stability. Pilots manually adjusted the aircraft's pitch, roll, and yaw to maintain flight. ••
- *Propulsion mechanism*: Small piston or internal combustion engines powered a propeller, producing thrust to overcome drag and maintain forward motion. These engines were manually started and mechanically simple, representing the earliest examples of motorized UAV propulsion. ••
- *Line-of-sight operation*: Drones were flown strictly within visual range, as there were no onboard sensors, telemetry, or navigation aids. The operator had to maintain direct visual contact throughout the mission for manual control and recovery. ••

2.4.1.2 Key components of first-generation drones

1. *Propeller*: Provided thrust for forward motion using a small piston engine.